IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

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Introduction

- SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upwards of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage. Therefore, if we can determine if the first stage will land, we can determine the cost of a launch.
- The objective of this project is to analyze a data set of SpaceX Falcon 9 launches since 2012, and to find a prediction model to classify launches outcome (success or failure)
- The questions I should answer are :

- Identify the relationships between the explaining variables such as the launch site, the payload mass, the orbit, the flight number and the launch outcome
- Create several kinds of visualization and insights to communicate the answers
- Find the best prediction model to classify the outcomes

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology
- Perform data wrangling
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models

Data Collection

Two different process were used to collect SpaceX launch data :

- API calls using python requests library
- Webscrapping using python BeautifulSoup library to gain information from the Wikipedia SpaceX launch page

API Calls

Webscrapping

Data Collection – SpaceX API

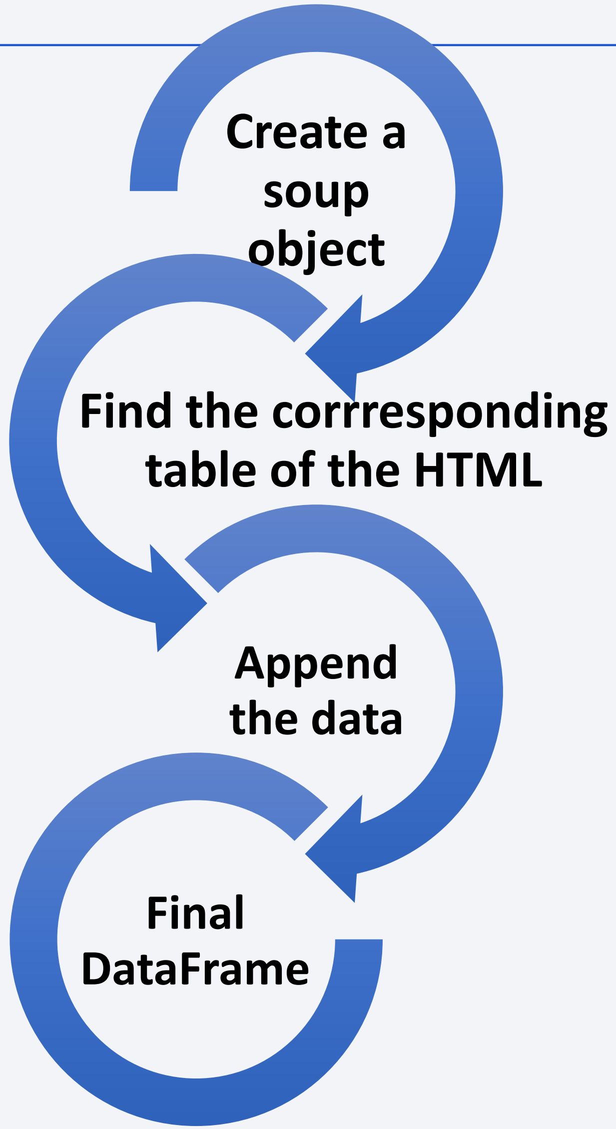
- I first collected data using python requests library and created a final DataFrame with SpaceX launches data

- Github URL of the completed SpaceX API calls notebook

Data Collection - Scraping

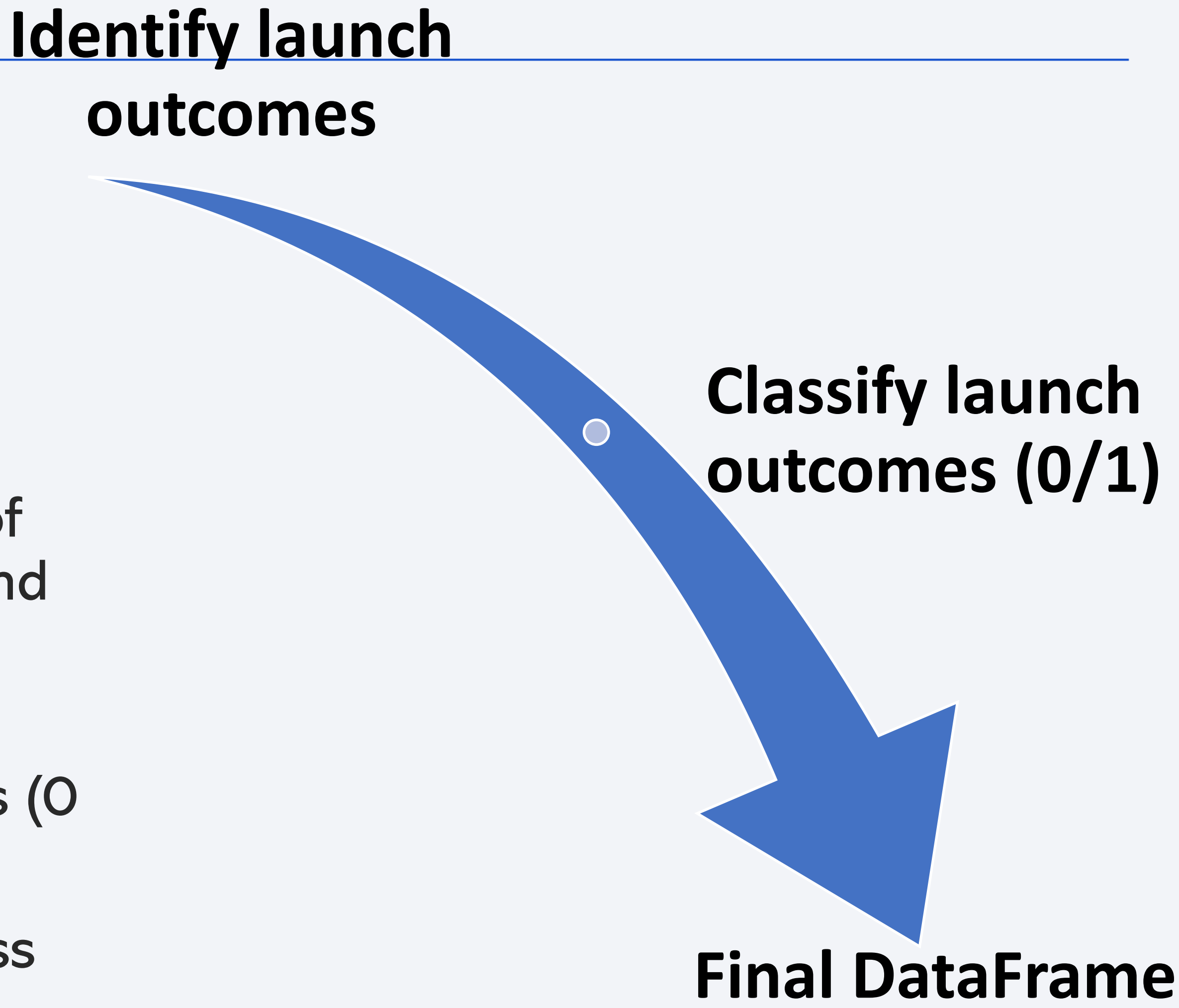
- The second method is to use the BeautifulSoup library to extract data from the Wikipédia’s SpaceX page.

- Github URL of the SpaceX data scraping



Data Wrangling

- I used the cleansed DataFrame I created during the scraping
- I identified the different launch outcomes and their count
- I created a class based on the outcome of the launch : 0 = unsuccessful outcome and 1 = Successful outcome
- I created a new column in the df called “landing class” with the outcome's values (0 or 1)
- GitHub URL of the data wrangling process



EDA with Data Visualization

- Vizualisation of Launch Site over Flight Number, Payload Mass over Launch Sites, Payload Mass over Orbits and Flight Numbers over Orbits to identify any relationship having an influence on classification
- Bar chart of Success Rates over Orbits to identify the better Orbit
- Line chart of succes rates over years to analyze the evolution of the launch outcomes
- Github URL

EDA with SQL

SQL queries I performed :

- Display the launch sites names
- Display the total payload mass carried by boosters launched by NASA
- Display the average payload mass carried by booster version F9 v1.1
- List the date when the first succesful landing outcome in ground pad was achieved
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List total number of successful and failure mission outcomes
- List all the booster_versions that have carried the maximum payload mass
- List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

• [GitHub URL]

Build an Interactive Map with Folium

- Map with marked launch sites for global visualization
- Markers of successful/failed outcomes with green color for success and red for failure in order to visualize the outcomes on a map
- Mouse Position in order to have the coordinates of each point
- Polylines between random Launch sites and its proximities (coast for example) to calculate distance between the sites and any point we choose.
- GitHub URL

Build a Dashboard with Plotly Dash

- Launch Site Drop-down list to visualize a pie chart of success rates of each site
- Payload Mass range slider to interactwith a scatter plot of Payload Mass over Class (success/failure) to identify which range of Payload Mass provides the best outcome
- GitHub URL

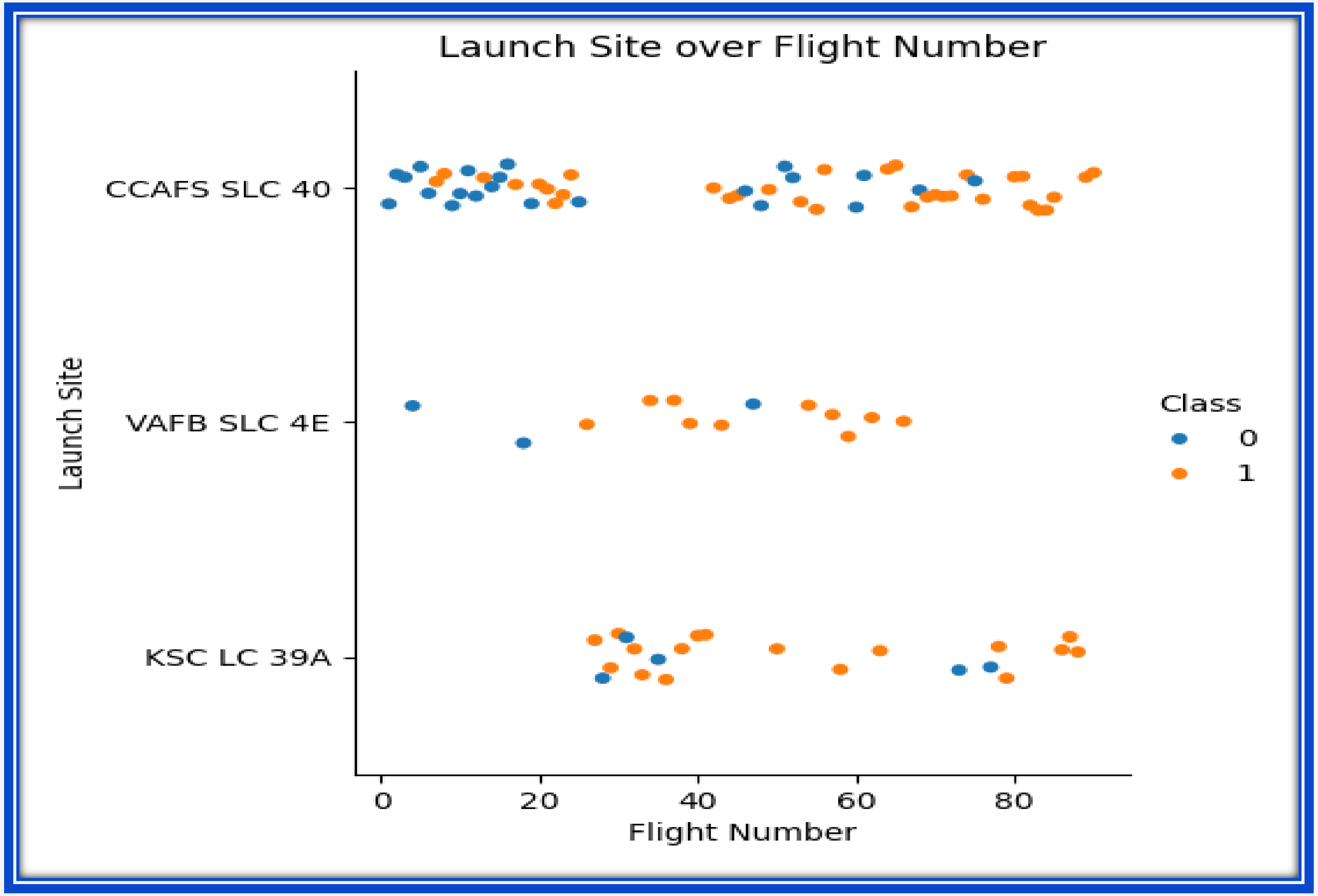
Predictive Analysis (Classification)

- I predicted launch outcomes using the same process for 3 different classification models (Logistic regression, SVM, Decision Tree) : data standardization, train/test/split, GridSearchCV using different parameters with a cross validation up to 10.
- Confusion matrix and accuracy score for each prediction model to identify True/False Positives/Negatives and the best model to choose
- Github URL

Section 2

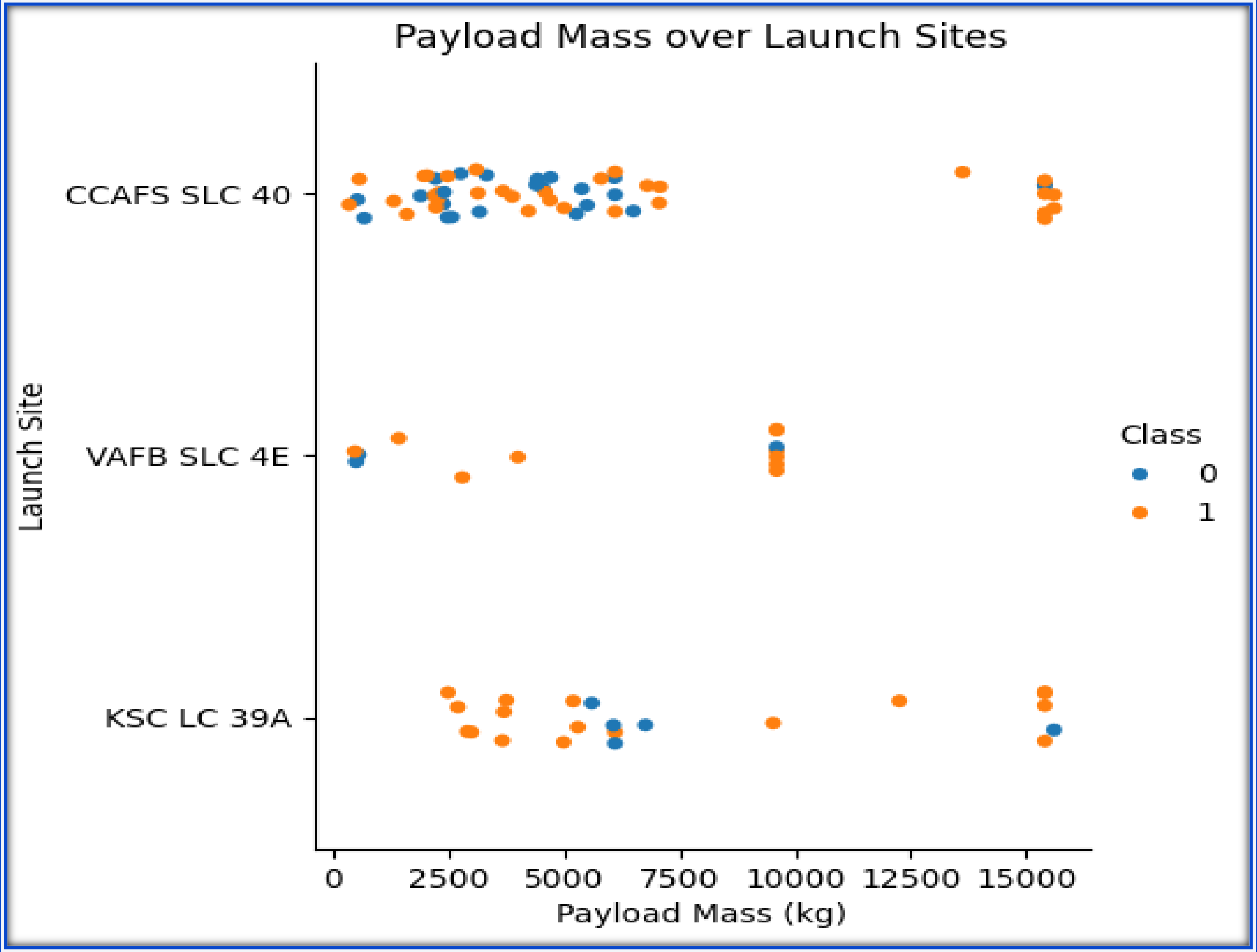
Insights drawn from EDA

Flight Number vs. Launch Site



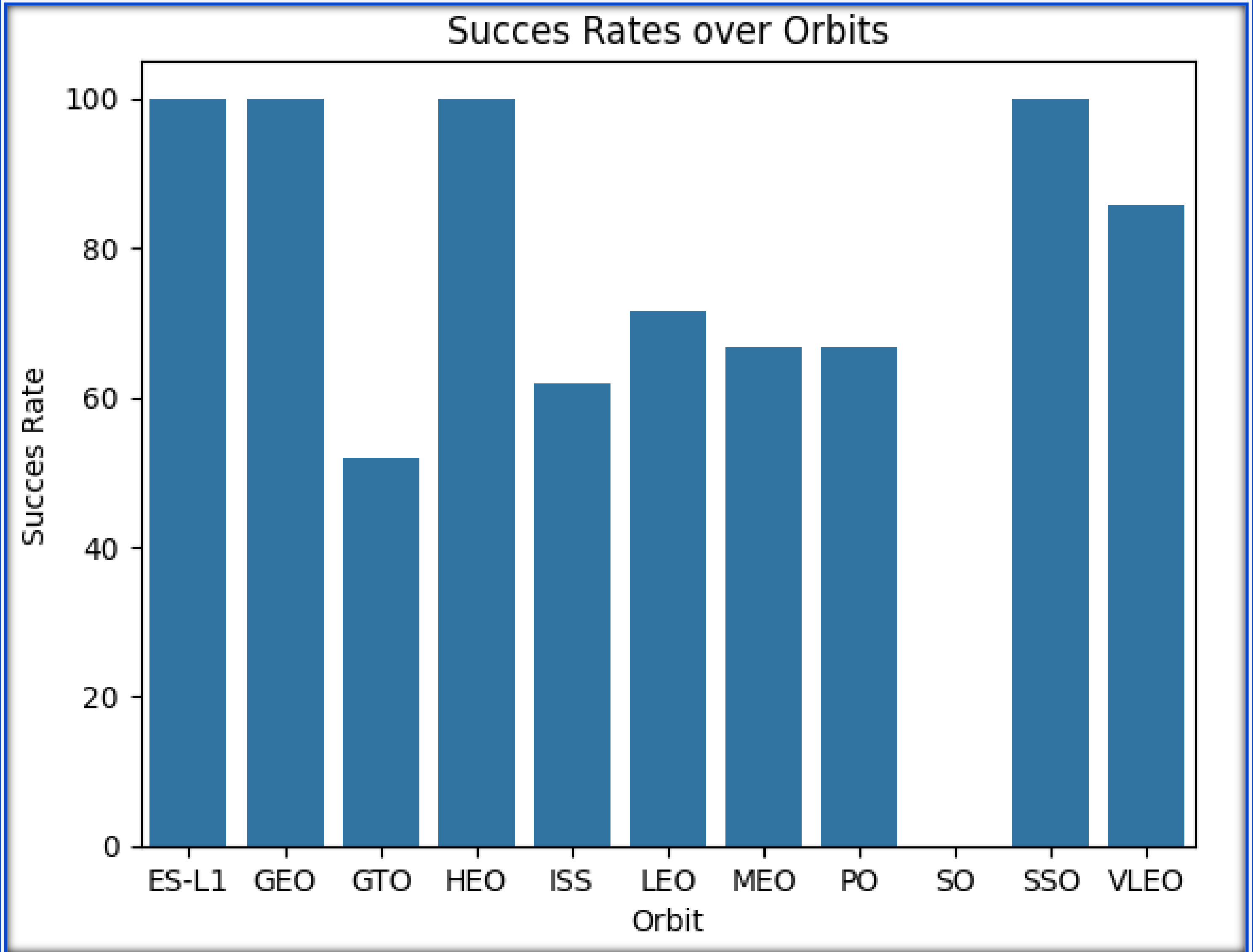
- CCAFS SLC 40 launch site has the most failure outcomes but also the most launch attempts.
- VAFB SLC 4E launch site has the highest outcome success rate but is also the least used.

Payload vs. Launch Site



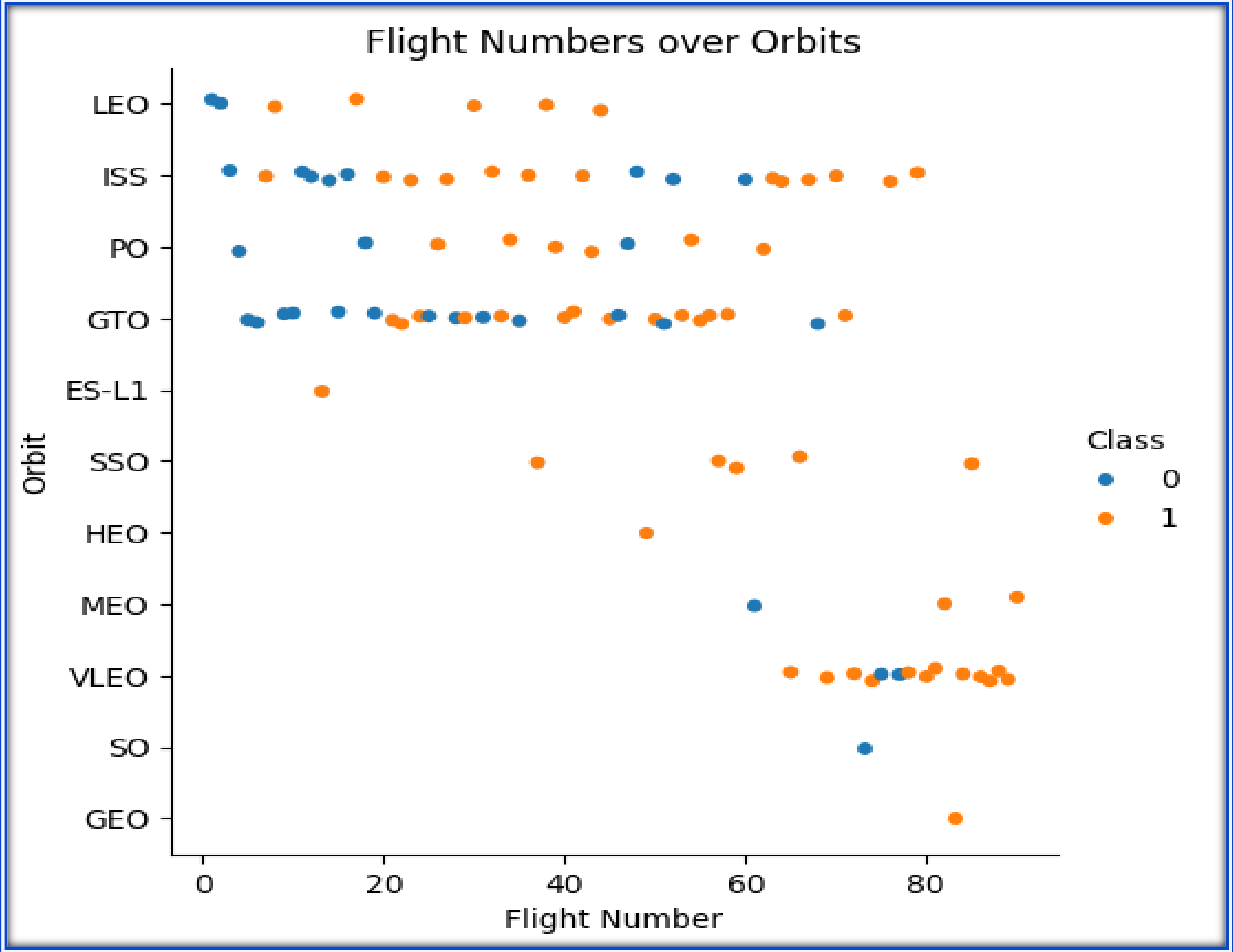
- CCAFS SLC 40 launch site has its best outcome success rate when the payload mass is about 15000
- There are no launches with a payload mass higher than 10000 for the VAFB SLC 4E launch site
- KSC LC 39 A launch site has the most regular payload mass use, with all the range from 0 to 15000 kgs used for the launches

Success Rate vs. Orbit Type



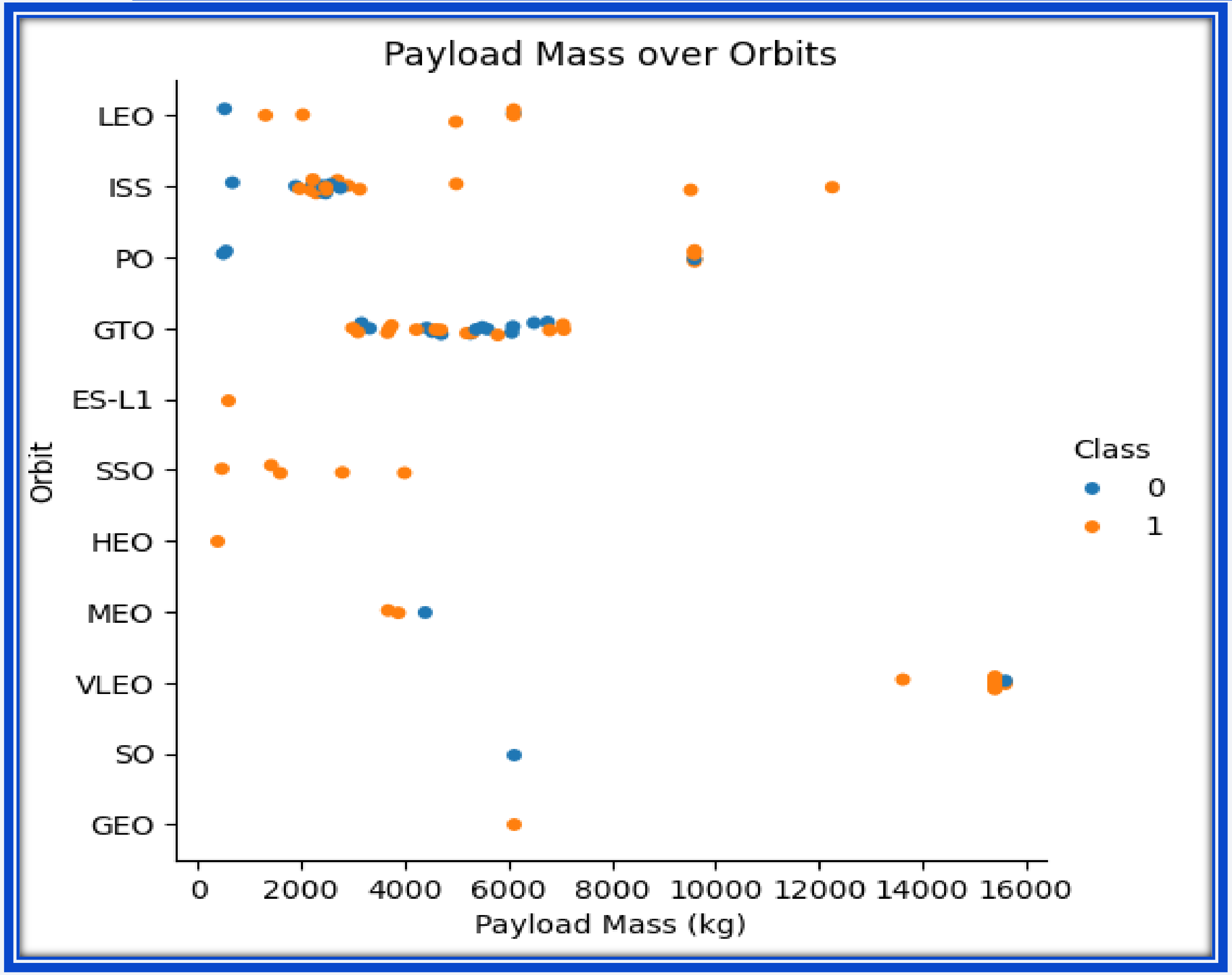
- ES-L1, GEO, HIEO, SSO have a 100% outcome success rate
- SO has the worst success rate, with 0 successful outcome

Flight Number vs. Orbit Type



- LEO, ISS, PO, GTO are the most used orbits
- GEO and SO are the less used orbits
- SSO, HEO, MEO and VLEO are used since the 40th flight, whereas LEO, ISS, PO, GTO are less used after it.

Payload vs. Orbit Type



- Most of orbits use a payload mass range between 0 and 7 000 kgs
- VLEO only launches flights with a payload mass higher than 13 000 kgs

Launch Success Yearly Trend

- Outcome success rate is significantly increasing since 2013
- A drop is visible between 2017 and 2018 with a decrease from 80% to 60%
- Last observed years shows a quite constant success rate (about 80%)

All Launch Site Names

There are 4 launch sites :

- CCAFS LC -40
- VAFB SLC -4E
- KSC LC -39A
- CCAFS SLC -40

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

5 records where launch sites begin with `CCA`

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- The total payload mass carried by boosters launched by NASA (CRS) is 45 596 kg

Customer	total_mass_kg
NASA (CRS)	45596

Average Payload Mass by F9 v1.1

Booster_Version	Average_payload_mass
F9 v1.1	2928.4
F9 v1.1 B1003	500.0
F9 v1.1 B1010	2216.0
F9 v1.1 B1011	4428.0
F9 v1.1 B1012	2395.0
F9 v1.1 B1013	570.0
F9 v1.1 B1014	4159.0
F9 v1.1 B1015	1898.0
F9 v1.1 B1016	4707.0
F9 v1.1 B1017	553.0
F9 v1.1 B1018	1952.0

- Booster version F9 v1.1 B1011 has the heaviest average payload
- Booster version F9 v1.1 B1003 has the lightest average payload

First Successful Ground Landing Date

- The first successful ground landing date is the 22nd of December 2015

Date	"Time"	Landing_Outcome
2015-12-22	Time	Success (ground pad)

Successful Drone Ship Landing with Payload between 4000 and 6000

- F9 FT B1022, F9 FT B1026, F9 FT B1021.2 and F9 FT B1031.2 are the boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Booster_Version	PAYLOAD_MASS__KG_	Landing_Outcome
F9 FT B1022	4696	Success (drone ship)
F9 FT B1026	4600	Success (drone ship)
F9 FT B1021.2	5300	Success (drone ship)
F9 FT B1031.2	5200	Success (drone ship)

Total Number of Successful and Failure Mission Outcomes

- There are 100 successful mission outcomes out of 101.

Mission_Outcome	COUNT("Mission_Outcome")
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

Booster_Version
F9 v1.0 B0003
F9 v1.0 B0004
F9 v1.0 B0005
F9 v1.0 B0006
F9 v1.0 B0007
F9 v1.1 B1003
F9 v1.1
F9 v1.1 B1011
F9 v1.1 B1010
F9 v1.1 B1012
F9 v1.1 B1013
F9 v1.1 B1014
F9 v1.1 B1015
F9 v1.1 B1016
F9 v1.1 B1018
F9 FT B1019
F9 v1.1 B1017
F9 FT B1020
F9 FT B1021.1
F9 FT B1022
F9 FT B1023.1
F9 FT B1024
F9 FT B1025.1
F9 FT B1026
F9 FT B1029.1
F9 FT B1031.1
F9 FT B1030
F9 FT B1021.2
F9 FT B1032.1
F9 FT B1034
F9 FT B1035.1
F9 FT B1029.2
F9 FT B1036.1
F9 FT B1037
F9 B4 B1039.1
F9 FT B1038.1
F9 B4 B1040.1
F9 B4 B1041.1
F9 FT B1031.2
F9 B4 B1042.1
F9 FT B1035.2
F9 FT B1036.2
F9 B4 B1043.1
F9 FT B1032.2
F9 FT B1038.2
F9 B4 B1044
F9 B4 B1041.2
F9 B4 B1039.2
F9 B4 B1045.1
F9 B5 B1046.1
F9 B4 B1043.2
F9 B4 B1040.2
F9 B4 B1045.2
F9 B5B1047.1
F9 B5B1048.1
F9 B5 B1046.2
F9 B5B1049.1
F9 B5 B1048.2
F9 B5 B1047.2
F9 B5 B1046.3
F9 B5B1050
F9 B5B1054
F9 B5 B1049.2
F9 B5 B1048.3
F9 B5B1051.1
F9 B5B1056.1
F9 B5 B1049.3
F9 B5 B1051.2
F9 B5 B1056.2
F9 B5 B1047.3
F9 B5 B1048.4
F9 B5B1059.1
F9 B5 B1056.3
F9 B5 B1049.4
F9 B5 B1046.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1059.2
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5B1058.1
F9 B5 B1049.5
F9 B5 B1059.3
F9 B5B1060.1
F9 B5 B1058.2
F9 B5 B1051.5
F9 B5 B1049.6
F9 B5 B1059.4
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5B1062.1
F9 B5B1061.1
F9 B5B1063.1
F9 B5 B1049.7
F9 B5 B1058.4

- These are all the boosters that carried maximum payload

2015 Launch Records

List of the failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015

- Months : January and April
- Booster version : F9 v1.1 B1012 and F9 v1.1 B1015
- Launch site : CCAFS LC-40

MONTH	Booster_Version	Launch_Site	Landing_Outcome
01	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

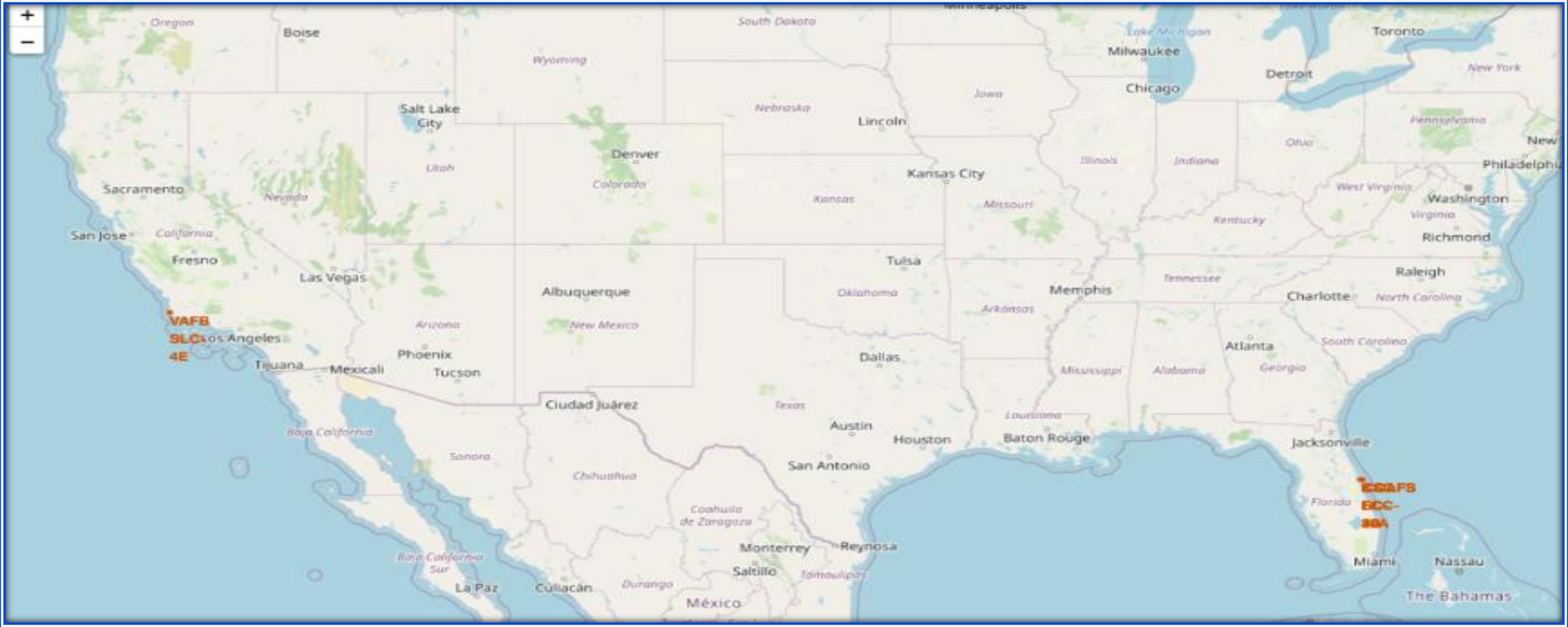
- Ranking of the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order :

Landing_Outcome	COUNT("Landing_Outcome")
Success (drone ship)	5
Success (ground pad)	3
Precluded (drone ship)	1
Failure (drone ship)	5
Controlled (ocean)	3
Uncontrolled (ocean)	2
No attempt	10
Failure (parachute)	2

Section 3

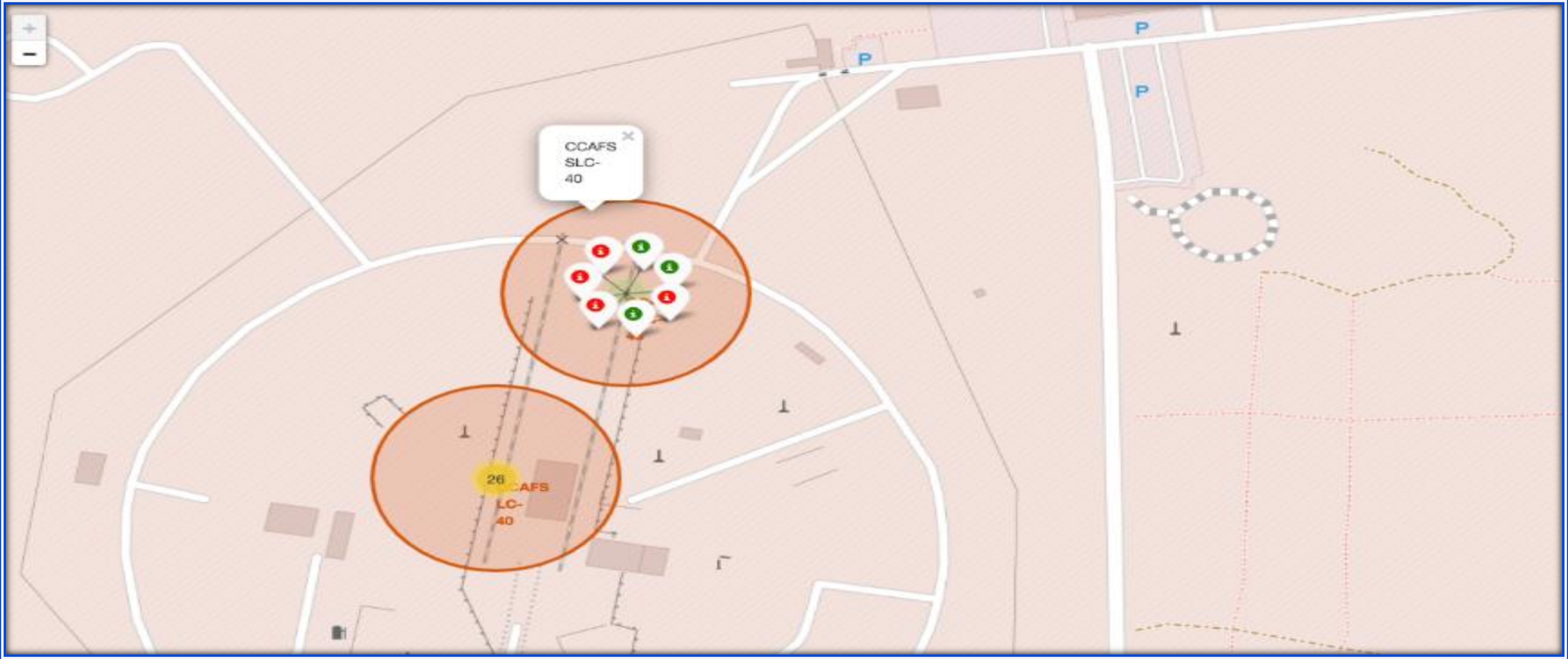
Launch Sites Proximities Analysis

Launch Sites localisation



- VAFB SLC 4E is located at the South-West near Los Angeles, and the CCAFS LC-40 and CCAFS SLC-40 are located at the Souht-East in Florida. All launch sites are located near to the coast.

Success/Failure Outcome geographical vizualization



- Successful launches are marked in green whereas unsuccessful ones are marked in red.

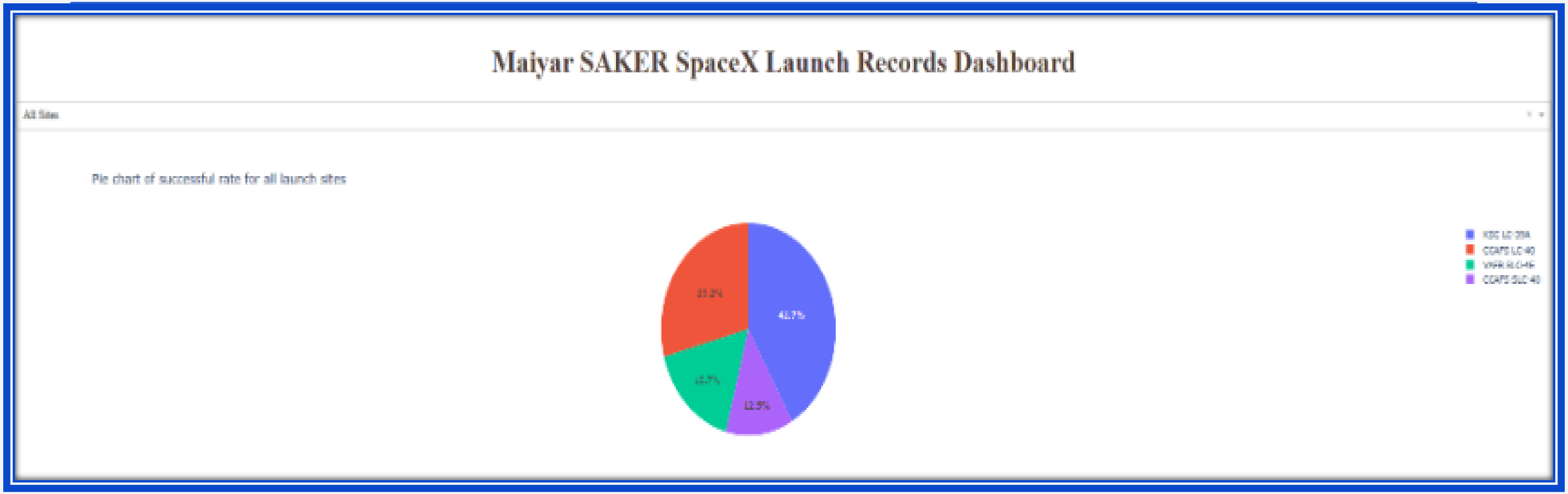
Distance between CCAFS SLC-40 Launch site and the coast

- CCAFS SLC –40 Launch Site is located 900 meters near to the coast

Section 4

Build a Dashboard with Plotly Dash

Launch success count dashboard

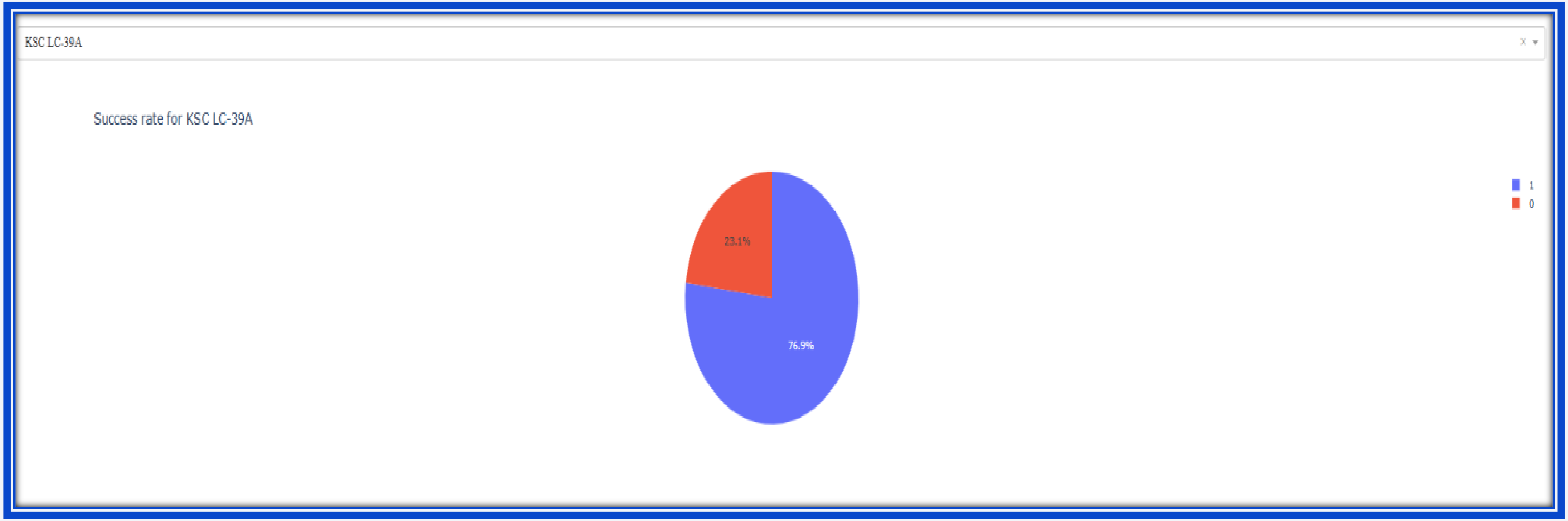


- This pie shows the proportion of successful rates over the launches :

- 41.7 % (the most) of the successful outcomes took place at the KSC LC -39 A launch site
- 12.5 % (the least) of the successful outcomes took place at the CCAFS SLC -40 launch site

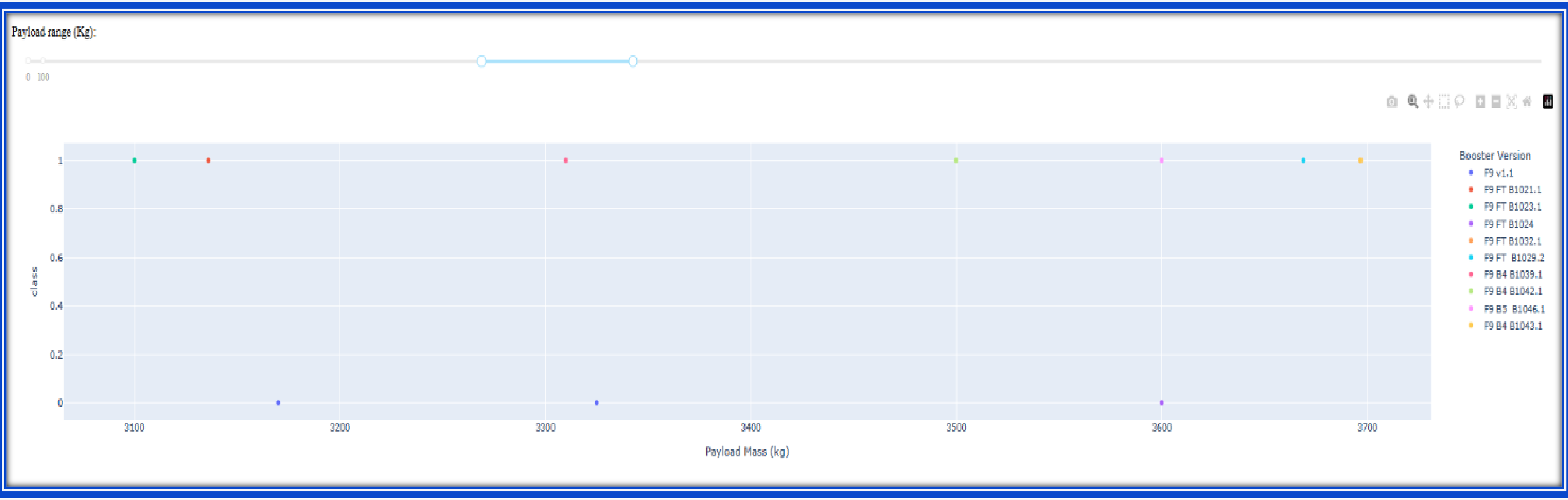
Highest Success Rate Launch Site

- KSC LC -39A has the highest launch success rate (77%)



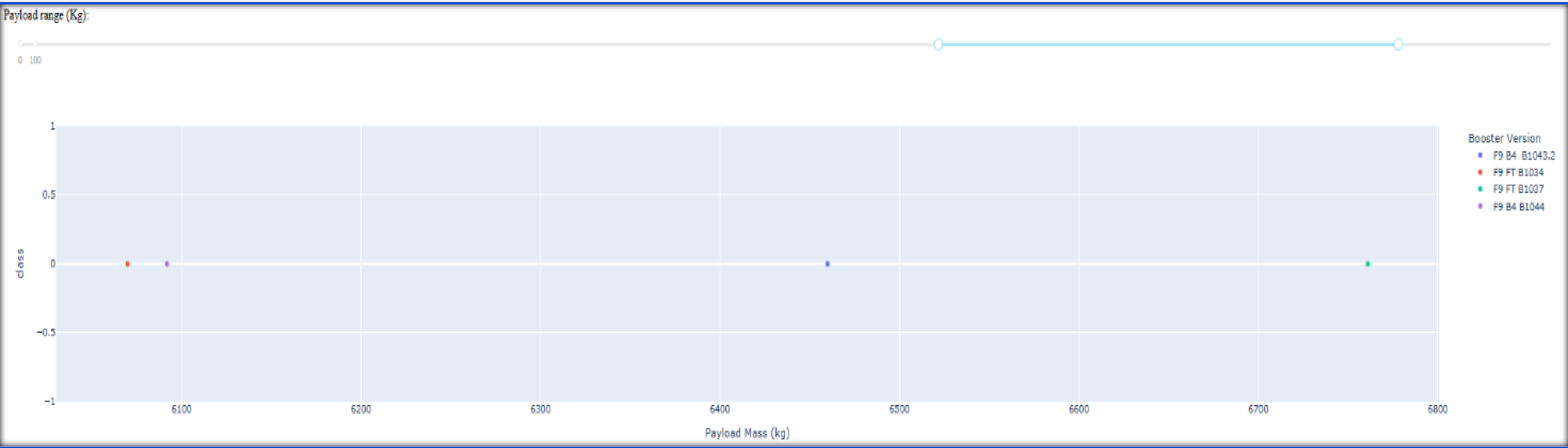
Highest Launch Success Rate Payload range

- Range between 3000 and 4000 kg has a success rate of 70% :

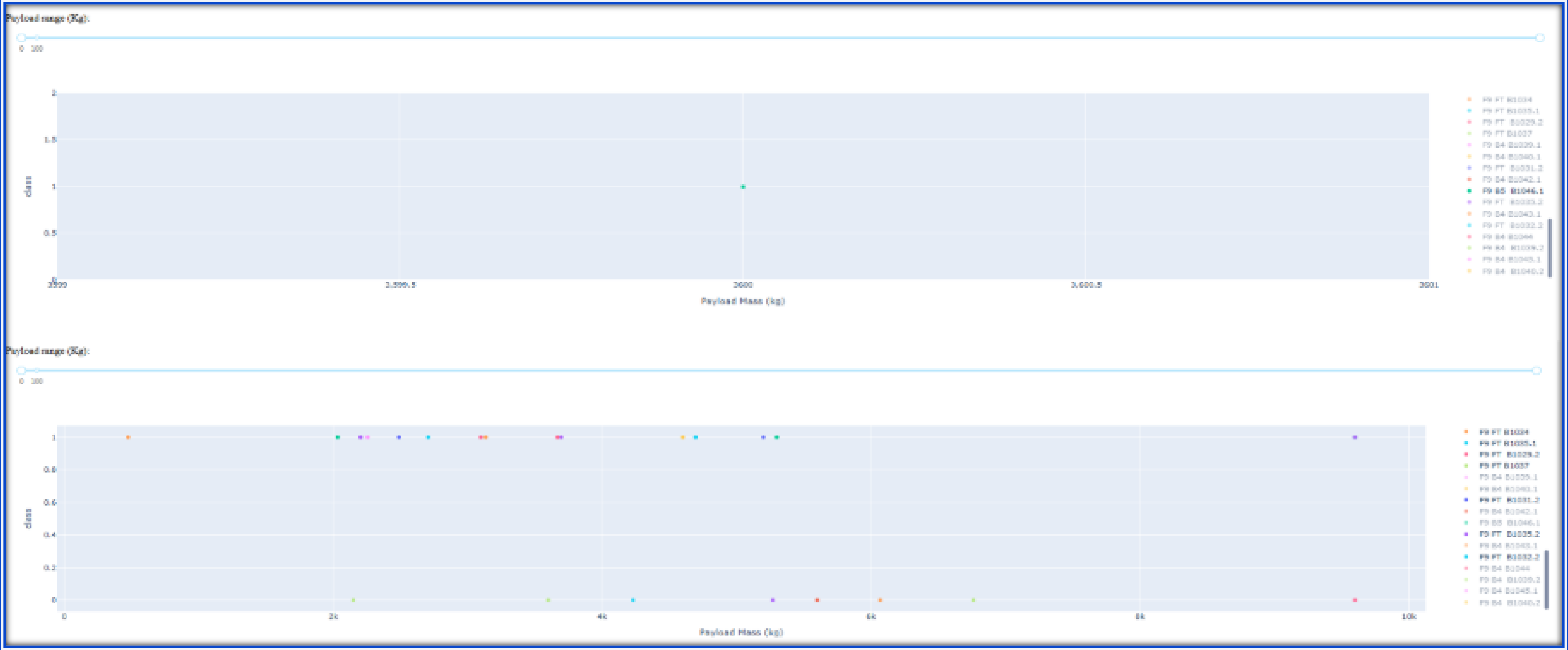


Lowest Launch Success Rate Payload range

- Between 6000 and 7000 kg, the success launch is 0% :



Booster Versions with the highest launch success rate



- F9 B5 has a succes rate of 100% but with only one shot, whereas F9 FT has a quite good success rate (around 63%) with a lot of launches :

Section 5

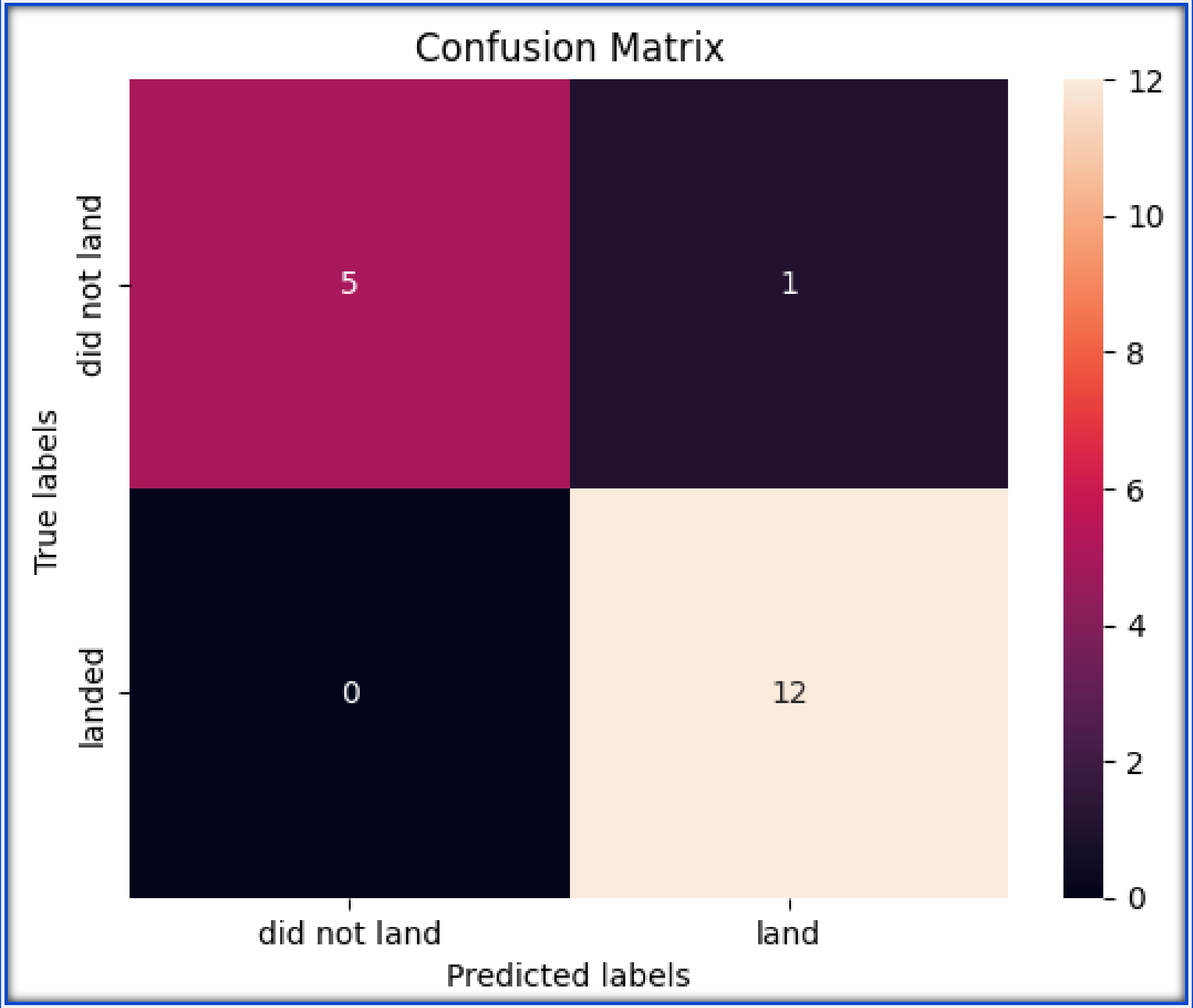
Predictive Analysis (Classification)

Classification Accuracy

- Logistic Regression, SVM and KNN have the same accuracy : 83%
- The Decision Tree Classifier has the best accuracy : 94%

```
Logistic Regresion score : 0.8333333333333334
SVM score : 0.8333333333333334
Decision Tree score : 0.9444444444444444
KNN score : 0.8333333333333334
```

Decision Tree confusion Matrix



- The Decision Tree confusion matrix shows a very strong predictive model, with only one False Positive predicted and no False Negative.
- It shows the strong accuracy of the model : 17 good predicted labes over 18, so 94% accurate.

Conclusions

- I analyzed and visualized the data to show the relationship between the explanatory variable using charts, graphs, rankings.
- I provided a Map to geographically identify the launch sites and visualize their location and the outcomes.
- I created an interactive dashboard to compare the outcome success rate for all launch sites and how the payload range can have an influence on the classification of an outcome
- Finally, I implemented a strong Decision Tree Classifier model to predict the outcome of the launches, with an accuracy of 94%.

Appendix

- Project Github

Thank you!